Testing for Racial Bias in Police Traffic Searches

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Abstract

I develop a framework to detect and measure racial bias in police traffic searches. Officers are evaluated individually, permitting unrestricted officer heterogeneity and nonrandom assignment of drivers to officers. By using a threshold model with random thresholds, the direction and intensity of bias can vary with the probability that a driver carries contraband. Sharp bounds on the intensity of bias are derived using bilinear programs. I evaluate 50 officers from the Metropolitan Nashville Police Department and find 6 officers to be biased. Estimates suggest that the intensity of bias varies with the probability that a driver carries contraband.

JEL codes: C21, C36, J15, K14, K42

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1 Introduction

Disparities across race, sex, and other protected classes arise in many settings, including the labor market (Card et al., 2016; Agan and Starr, 2018; Kline et al., 2022), the criminal justice system (Arnold et al., 2018; Feigenberg and Miller, 2022), healthcare (Obermeyer et al., 2019; Wasserman, 2023), lending markets (Dobbie et al., 2021; Bhutta and Hizmo, 2021; Bartlett et al., 2022), among others. However, measuring the extent to which bias contributes to the disparities, if at all, is difficult due to methodological challenges and data limitations.

In this paper, I develop a framework to test for and measure bias, and I apply this framework to study racial bias in police traffic searches. I employ a threshold model where the officer searches a driver only if the driver's probability of carrying contraband ("risk") exceeds a threshold. This threshold represents the officer's preference for searching drivers, as well as other factors that influence the search decision. The proposed test differs from existing methods in four ways.

First, whereas earlier papers assume fixed thresholds, I allow thresholds to be random. This permits a richer form of bias where the direction and intensity of bias depends on the risk of the driver. For example, biased officers may search white drivers with 10% risk 20% of the time, but search equally risky black drivers 40% of the time, and this disparity can vary across risk. A biased officer may even search whites more than blacks at one level of risk, and search blacks more than whites at another level of risk. I show how the dependence between bias and risk may be learned despite how risk is unobserved by the researcher and cannot be credibly estimated. Understanding this dependence can aid the development of anti-bias training by identifying subgroups of drivers most affected by bias.

Second, I employ a partial identification approach since random thresholds preclude point identification. Testing for bias entails testing whether the sharp identified set for the distributions of officer thresholds (i.e., the smallest set of distributions consistent with the model and data) includes an equivalent pair of distributions for white and minority drivers. If not, then the officer's thresholds must differ by race, implying bias. The intensity of bias may be inferred from how dissimilar the distributions of thresholds are across race. Identification accounts for the endogeneity in search decisions, thus circumventing inframarginality bias. I utilize bilinear programs (BPs), which resemble linear programs but also include bilinear terms (i.e., the product of two distinct variables in the BP).[1] Restrictions on the model can be layered in a transparent manner as constraints to the program.

Third, I use an alternative instrumental variables (IV) strategy to aid in identification. The marginal outcome test popularized by Arnold et al. (2018) relies on instruments shifting the thresholds of decision makers while ensuring all decision makers face the same distribution of risk (e.g., random assignment of judges to defendants). Such instruments are difficult to justify in the context of police traffic searches because officers select their own distributions of risk by selecting whom to stop. Conversely, I employ an IV that shifts the distribution of risk among drivers stopped without shifting the officer's threshold. For each race of drivers, this generates a sequence of data points that must be consistent with a single distribution of thresholds, thereby constraining what the distribution can be. This is similar to how an IV identifies demand functions: the instrument shifts supply without shifting demand, generating a sequence of equilibria tracing out the demand curve.

Fourth, whereas existing tests analyze officers jointly and either explicitly or implicitly compare search decisions across officers, the proposed methods are applied to each officer separately. This is possible since my IV is able to vary the risk of drivers stopped for each officer separately. This permits unrestricted heterogeneity across officers in their distributions of thresholds and risk.

I apply the proposed methods on a panel data set tracking officers in the Metropolitan Nashville Police Department (MNPD) between 2010 and 2019. Since I evaluate each officer

[1]The bilinear terms imply a non-convex optimization problem, yet BPs can be solved to provable global optimality. BPs are not only novel in the context of discrimination, but also in the context of partial identification and econometrics in general.

separately, I restrict my attention to the 50 officers with the most number of searches. On average, these officers have made over 2,100 stops and 250 searches for each group of drivers, and account for one third of all searches in the data. I use the time and day of the traffic stop as the instrument, controlling for driver and neighborhood characteristics. Across two sets of estimates, six officers fail the test at the 5% significance level. For each of these officers, I estimate bounds on the average intensity of bias, as well as how the intensity of bias varies with the risk of the driver.

The paper proceeds as follows. Section 2 reviews the literature on testing for racial bias; Section 3 presents the model of an individual officer's search decision; Section 4 formalizes how bias may be detected and measured; Section 5 discusses the application; and Section 6 concludes.

2 Literature review

It is well documented that black civilians are more likely to be stopped, searched, and killed by police officers compared to white civilians (Gelman et al., 2007; Pierson et al., 2020).[2] However, it is challenging to determine the extent to which these disparities stem from racial bias because researchers have limited information on civilians, officers, and their interactions. In this section, I summarize earlier approaches to detecting racial bias.

Knowles et al. (2001) lay the foundation for detecting racial bias in traffic searches by operationalizing the outcome test proposed by Becker (1957, 1993). Officers are modeled as being homogeneous and only search drivers whose risk of carrying contraband exceeds a threshold.[3] Bias is defined as a racial disparity in thresholds. The researcher's objective is thus to recover the thresholds for black and white drivers.

If risk is observed by the researcher and continuously distributed over the unit interval, then the thresholds are identified from the risk of the white and minority drivers at the margin

[2] See also the Fatal Force database by the Washington Post.

[3] Persico and Todd (2006) generalize the model to allow heterogeneity across officers.

of search. However, risk is unobserved. To recover the threshold, Knowles et al. (2001) use an equilibrium model where heterogeneous drivers decide whether to carry contraband in response to the probability they are searched.[4] In equilibrium, drivers carry contraband with probability equal to a fixed threshold they face given their race.[5] This results in a straightforward test for bias: if officers have different success ("hit") rates conditional on searching white and minority drivers, then officers are biased.

Anwar and Fang (2006) propose an alternative test that allows for heterogeneity in officer decisions and driver risk. The authors allow different officers to have different thresholds and test for bias using pairwise comparisons of search decisions across groups of officers (e.g., white officers versus black officers). If both groups of officers are unbiased, then the ranking of their search rates should be the same regardless of the race of the driver. While this approach can detect bias, it cannot determine which group of officers is biased, nor which group of drivers is being discriminated against.

Arnold et al. (2018) made a significant contribution to the literature by using random assignment of defendants to judges as an instrument to detect racial bias in bail decisions. The authors assume thresholds are distributed continuously across decision makers. Under restrictions formalized by Canay et al. (2023), the thresholds of all decision makers can be point identified using the marginal treatment effect framework of Heckman and Vytlacil (2005). These restrictions include decision makers facing identical distributions of risk (hence the importance of random assignment) and modeling decision makers using the Extended Roy Model (i.e., fixed thresholds). This method is referred to as the marginal outcome test.

Using data from Florida and Texas, Gelbach (2021) shows the implications of the marginal outcome test are violated in the context of police traffic searches. The author points to

[4] Recent tests of bias, including mine, assume decision makers operate independently. See Persico (2002); Persico and Todd (2006) for equilibrium models where officers coordinate.

[5] The argument is that drivers who are more likely to carry contraband will be searched more frequently. These drivers are therefore discouraged from carrying contraband. In equilibrium, all drivers of the same race carry contraband with equal probability and officers search each race at random.

different distributions of risk across officers as a potential reason.[6] Such differences can arise if officers are not randomly assigned to drivers or vary in their ability to assess the risk of drivers. Papers using the marginal outcome test to study bias in policing therefore require restrictions on the distributions of risk. For example, Marx (2022) requires the distributions of risk to be common across officers. Feigenberg and Miller (2022) allow the distributions of risk to vary across officers, but rule out sample selection on unobservables.[7,8]

Other papers have used statistical approaches to test whether civilian race has an effect on police decisions, including stop-and-frisk and use of force (Ridgeway, 2006; Grogger and Ridgeway, 2006; Gelman et al., 2007; Ridgeway and MacDonald, 2009; Goel et al., 2016a,b; Fryer Jr, 2019; MacDonald and Fagan, 2019; Knox et al., 2020; Gaebler et al., 2022). These papers either assume that the distribution of risk may be balanced across races, or cannot attribute the effect of race to racial bias. Knox et al. (2020) is noteworthy for emphasizing the difficulty of identifying the effect of race on post-stop decisions alone (e.g., use of force, traffic searches) due to sample selection.[9]

[6] Frandsen et al. (2023) and Sigstad (2026) propose tests for the IV monotonicity condition implied by the marginal outcome test framework. In a panel of Brazilian judges, Sigstad (2026) finds that monotonicity is violated.

[7] The difference-in-differences strategy used by Goncalves and Mello (2021) to study racial bias among officers writing speeding tickets also rules out sample selection on unobservables.

[8] Arnold et al. (2022) allow decision makers to face different distributions of risk, but require parametric assumptions on the joint distribution of thresholds and risk. Simoiu et al. (2017), Pierson et al. (2020) Chan et al. (2022) impose similar parametric assumptions to identify thresholds.

[9] The authors show that, under a principal strata framework, identifying the effect of race on post-stop decisions is only possible in the knife-edge scenario where the biases from sample selection and omitted variables cancel each other out.

3 Model

In this section I model the search decision of a single officer (he) for drivers who are stopped (she).[10] Since officers are analyzed individually, I suppress the officer index. The analysis is conditional on stopped drivers and their observable characteristics. I suppress the notation for this conditioning for brevity.

3.1 Setup and notation

For each stop i, the officer observes the driver's race $R_i \in \{w, m\}$ (white or minority), and a set of non-race characteristics $V_i \in \mathcal{V}$ such as the driver's demeanor.[11] The econometrician only observes R_i but not V_i. Characteristics of the driver and the stop observed by the econometrician are implicitly conditioned on throughout.

The driver may or may not carry contraband (e.g., drugs, weapons), denoted by $Guilty_i \in \{0, 1\}$. The officer does not know whether the driver is guilty unless he performs a traffic search, denoted by $Search_i \in \{0, 1\}$. At the end of each traffic stop, the officer reports in the data whether a search was conducted and whether a "hit" occurred, i.e., contraband was found,

$$Hit_i \equiv Search_i \times Guilty_i.$$

I assume that the officer finds contraband if and only if he searches a guilty driver, as in Knowles et al. (2001) and Anwar and Fang (2006).

Drivers are drawn from a distribution that depends on the setting of the stop, $Z_i \in \mathcal{Z}$. For example, if Z_i is the hour and day of the stop, and the composition of drivers on the

[10]If officers work in pairs, then the search decision corresponds to a pair of officers, and pairs of officers are assumed to be fixed across stops.

[11]Components of V_i may be observed by the officer prior to the stop. Some components of V_i may also be observable to one officer but not another, which allows different officers to form different assessments of the driver's risk.

road changes with time, then different types of drivers will be stopped across different values of Z_i.[12] The setting is observed by both the officer and econometrician and will play the role of an instrument.

The officer's search decision may be written as

$$Search_i \equiv \mathbb{1}\left\{G(R_i, V_i, Z_i) \geq T_i\right\}, \tag{1}$$

where

$$G(r, v, z) \equiv \mathbb{P}\{Guilty_i = 1 \mid R_i = r, V_i = v, Z_i = z\}$$

is the probability that the driver carries contraband, which I refer to as the "risk" of the driver; and T_i is a random threshold that may be interpreted as the officer's perceived cost of searching a driver. The risk and threshold can depend on observable characteristics of the driver. This model is a relaxation of the Extended Roy Model proposed by Canay et al. (2023) (henceforth, CMM) as I allow T_i to be random even after conditioning on all characteristics of the driver observable to the officer.[13,14] This allows the model to reflect how an officer's threshold may vary for reasons other than the driver (e.g., the officer may receive shocks to his motivation and risk aversion). This also permits a richer notion of bias

[12]If there are variables that inform the officer's stop decision and are visible for some values of Z_i but not others, then the distribution of drivers stopped will vary with Z_i even if the composition of drivers on the road do not. Such variation is used in the Veil of Darkness test by Grogger and Ridgeway (2006) to test whether race affects the stop decision.

[13]T_i can be written as $T_i = t(R_i) + \varepsilon_i$, where $t(\cdot)$ is a deterministic function and ε_i is a shock that may depend on race. The standard model with fixed thresholds corresponds to the case where $\varepsilon_i = 0$. See Online Appendix A.1 to see how the model in CMM may be extended to have a random threshold.

[14]The judge IV is similar to a random threshold insofar as there is a distribution of thresholds across judges. The judge IV allows the researcher to make incremental changes to the threshold, which is crucial to the identification results in CMM. However, in my setting, it is not possible for the researcher to make incremental changes to an officer's threshold.

where the intensity and direction of bias can vary with the risk of the driver.

To derive a logically valid test for bias, I assume the following.

Assumption 1.

(i) $T_i \mid R_i = r$ *is identically distributed across stops* i *for* $r \in \{w, m\}$.

(ii) $T_i \perp\!\!\!\perp (Guilty_i, Z_i, V_i) \mid R_i = r$ *for* $r \in \{w, m\}$.

Assumption 1(i) allows me to pool observations within driver race to infer an officer's thresholds. Assumption 1(ii) states that, conditional on driver race R_i, the officer's threshold is jointly independent of the guilt of the driver $Guilty_i$, the setting Z_i, and driver characteristics, V_i.[15]

Let F_X denote the CDF for an arbitrary random variable X. Under (1) and Assumption 1, the probability that a driver is searched is

$$
\begin{aligned}
&\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z, V_i = v\} \\
&= \mathbb{P}\{G(R_i, V_i, Z_i) \geq T_i \mid R_i = r, Z_i = z, V_i = v\} \\
&= \mathbb{P}\{G(r, v, z) \geq T_i \mid R_i = r, Z_i = z, V_i = v\} \\
&= \mathbb{P}\{G(r, v, z) \geq T_i \mid R_i = r\} \\
&= F_{T|R}(G(r, v, z) \mid r),
\end{aligned}
$$

where the third equality follows from Assumption 1(ii). The probability a driver is searched is equal to the probability the officer's threshold falls below the driver's risk, which implies

[15] Most existing tests for bias, including the one I propose, are intended for an "essentialist" perspective of race, under which race is a fixed biological characteristic. Under a "constructivist" perspective, race is instead a function of the physical and contextual features of the driver, e.g., $R_i = r(V_i, Z_i)$ for some function r (Rose, 2023). This poses challenges for measuring racial bias as a change in race necessarily coincides with changes in other driver characteristics. If the characteristics determining race are independent of the threshold T_i, then the methods I propose may still be used to detect bias. Otherwise, any "bias" detected should be interpreted as disparate impact.

riskier drivers are more likely to be searched. This leads to the following definition of bias.

Definition 1.

(i) *The officer is racially biased if* $F_{T|R}(\cdot \mid w) \neq F_{T|R}(\cdot \mid m)$.

(ii) *The officer is racially biased at risk* $g \in [0,1]$ *if*

$$\beta(g) \equiv F_{T|R}(g \mid m) - F_{T|R}(g \mid w) \neq 0,$$

where $\beta(g)$ *measures the intensity of bias at risk* g. *If* $\beta(g) > 0$ *(*$\beta(g) < 0$*), then the officer is biased against minority (white) drivers with risk* g.

Definition 1 is similar to racial τ-bias proposed by CMM and corresponds to disparate treatment of equally risky drivers of different race. Since $\beta(g)$ can vary with g and change sign, the intensity and direction of bias can vary with the risk of the driver, allowing for a more nuanced analysis of bias. This feature of the model arises from the random threshold and distinguishes my model from earlier models where, conditional on race and risk, an officer searches all drivers or none at all.[16]

3.2 Discussion

Assumption 1(ii) is crucial to identification and warrants careful discussion. I also discuss how to select an instrument, and how the methods extend to other settings.

3.2.1 Exogeneity assumption

There are three independence conditions in Assumption 1(ii): (i) $T_i \perp\!\!\!\perp \mathit{Guilty}_i \mid R_i$; (ii) $T_i \perp\!\!\!\perp Z_i \mid R_i$; (iii) $T_i \perp\!\!\!\perp V_i \mid R_i$. The first condition simply restricts the officer to infer the probability a driver carries contraband using only details he observes during the traffic

[16]Fixed thresholds are nested in my framework and correspond to a degenerate distribution of T_i. Whether thresholds are fixed or random, bias may be thought of as different distributions of thresholds across races.

stop—R_i, V_i, and Z_i—and not from his thresholds. The second condition relates to the instrument, and I discuss it in the next section dedicated to the instrument.

I focus my discussion here on the third independence condition, which ensures that V_i influences the search decision exclusively through the driver's risk and not through the threshold, and is implicitly assumed in existing fixed-threshold models. The Extended Roy Model of CMM imposes a similar restriction, where only one side of the selection equation may depend on variables unobserved by the econometrician. Without such an assumption, it is impossible to distinguish between differences in V_i across races from racial bias (Canay et al., 2023).[17]

How well $T_i \perp\!\!\!\perp V_i \mid R_i$ is supported depends on the interpretation of T_i and the richness of the data. Following CMM, I show in Online Appendix A.1 that the threshold can be decomposed as $T_i = B_i + C_i + \lambda_i$, where B_i reflects the officer's taste for searching drivers, C_i reflects other considerations made by the officer (e.g., safety),[18] and λ_i reflects the error in assessing a driver's risk. Supporting $T_i \perp\!\!\!\perp V_i \mid R_i$ amounts to supporting $(B_i, C_i, \lambda_i) \perp\!\!\!\perp V_i \mid R_i$.One approach is to simplify the interpretation of T_i by assuming away its components. For example, Knowles et al. (2001), Anwar and Fang (2006), and Persico (2009) assume $\lambda_i = 0$. Another approach—which I employ in the application—is to invoke a conditional independence assumption where V_i and T_i are independent conditional on an appropriate set of covariates. The success of this second approach depends on the credibility of the restrictions and the richness of the data.

For instance, suppose officers observe a driver's criminal history and receive additional utility when searching drivers with criminal histories. This implies different distributions of B_i for drivers with and without criminal histories. If a driver's criminal history is not observed by the econometrician, then it becomes a component of V_i and Assumption 1(ii)

[17]CMM primarily focus their discussion on the marginal outcome test of Arnold et al. (2018). The random threshold precludes the use of the marginal outcome test as a means to test for bias since there is no longer a single value of risk associated with a driver at the margin of search for each race.

[18]Kleinberg et al. (2018) refer to this as omitted payoff bias.

is violated. This invalidates the test since racial disparities in the distribution of T_i may potentially be explained by racial disparities in criminal histories rather than bias. This issue may be addressed by conditioning the analyses on drivers' criminal histories, as in Feigenberg and Miller (2022), as well as other factors that may influence an officer's search preference (e.g., non-race driver demographics).

Officers may also consider factors besides their taste for searching vehicles, which are captured by C_i. For example, officers may consider the opportunity cost to searching a vehicle, such as the time spent on other police calls. A potential concern is that minority drivers are more likely to be stopped in areas with greater need of policing compared to white drivers, leading to a higher opportunity cost of searching minority drivers and creating racial disparities in C_i. Disparities in T_i may therefore be driven by differences in demands for policing instead of bias. To avoid this issue, the analysis should condition on variables correlated with R_i and C_i, such as the volume of calls for police services.

Finally, officers may make errors when assessing the risk of the driver, which are contained in λ_i. Separating λ_i from the other components in T_i is difficult and requires direct measures of officer beliefs (Bohren et al., 2019, 2023).[19] If measurement error exists but is assumed to be idiosyncratic, then the proposed methods remain valid, although any bias detected cannot be interpreted as being taste-based. But if officers are assumed to be better at inferring the risk of drivers who are (for example) nervous as opposed to calm, and the demeanor of the driver is contained in V_i, then $\lambda_i \not\perp\!\!\!\perp V_i \mid R_i$ and Assumption 1(ii) is violated.

Satisfying $T_i \perp\!\!\!\perp V_i \mid R_i$ can therefore be a challenge. These challenges are not specific to my methodology or setting, but extend to earlier methods and other settings where there may be multiple determinants to the decision maker's threshold. Supporting Assumption 1 thus requires careful economic reasoning and sufficiently rich data.

[19] Alternatively, one can assume that B_i and C_i are fixed conditional on race, and $\mathbb{E}[\lambda_i \mid R_i = r] = 0$ for $r \in \{w, m\}$. All variation in T_i therefore stems from λ_i.

3.2.2 Choosing instrumental variables

In an experimental setting, the distribution of an officer's threshold can be identified by exposing him to drivers with various levels of risk, and then measuring the probability of searches conditional on race and risk. Bias may then be tested for and measured by comparing the distributions of thresholds across race. The IV attempts to replicate this experiment by varying the risk of the drivers without varying the thresholds. This requires the instrument satisfy a relevance condition, $Z_i \not\perp\!\!\!\perp V_i \mid R_i$, and an exogeneity condition, $T_i \perp\!\!\!\perp Z_i \mid R_i$.[20] Since risk is never observed by the econometrician and the guilt status of drivers is only revealed for those who are searched, the distribution of thresholds can only be partially identified.

The intuition for partial identification is similar to that of using an IV to identify a demand curve, where the instrument exclusively shifts the supply curve to trace out the demand curve. In my setting, Z_i exclusively shifts the distribution of risk through shifting V_i, generating a sequence of search and hit rates that constrain what the distributions of T_i could be. Greater variation in risk induced by Z_i imposes stronger constraints. Yet, the relevance condition is not necessary to detect bias. In Section 4, I show an example where the proposed methods are able to detect bias even when the IV is irrelevant.

The identification argument neither restricts how V_i varies across race, nor how $G(R_i, V_i, Z_i)$ depends on V_i for either race. Identification is then attainable regardless of how differently $G(R_i, V_i, Z_i)$ is distributed for each race of drivers, and is robust to sample selection in traffic stops and statistical discrimination.[21] The identification argument also does not restrict how T_i or V_i varies across officers. This is because the variation in $G(R_i, V_i, Z_i)$ through Z_i *within* an officer is informative of the officer's thresholds. This paper thus offers a new test for bias in settings where decision makers are either exogenously

[20] The observed covariates that are implicitly conditioned do not have to satisfy these conditions.

[21] G_i may be distributed differently across races if there is sample selection, where officers stop white and minority drivers with different V_i. G_i may also be distributed differently across races if there is statistical discrimination in the sense of Aigner and Cain (1977), where G_i depends on V_i differently across races.

or endogenously exposed to different distributions of individuals, and does so by providing a method to analyze each decision maker separately.

There are two approaches to selecting an instrument. The first considers variables that are related to risk but independent of the threshold. For example, traffic diversions force drivers to deviate from their usual routes. This can change the composition of drivers in a police precinct, thereby changing the distribution of risk faced by officers of that precinct. If the traffic diversion stems from a road closure or traffic accident, then it may be reasonable to assume that T_i and its components are unaffected by the diversion. While data on certain forms of traffic diversions are available (e.g., road closures), the variation in risk they generate may be too small to effectively detect racial bias in police traffic searches. For this reason, I do not use traffic diversions as the instrument in the application.

The second approach to finding an instrument is to invoke a conditional independence assumption, where the dependence between T_i and Z_i can be broken by conditioning on covariates. I employ this approach in the application by choosing Z_i to be the time and day of the stop and controlling for factors related to safety. I assume that an officer's thresholds vary with Z_i to the extent that his safety varies with the time and day of a stop. Then conditional on these factors, the officer is equally willing to search drivers across different times and days.

As with conventional instruments, the primary challenge of choosing an IV is arguing that Z_i satisfies the exogeneity condition $Z_i \perp\!\!\!\perp T_i \mid R_i$. For instance, in the first example above where traffic diversions are used as an instrument, not all traffic diversions will be suitable instruments. Traffic diversions caused by a police investigation or a crime likely encourage officers to search more frequently. Such traffic diversions should then be excluded as instruments.

The second approach to choosing an instrument faces the same challenge. For instance, if Z_i is the time and day of stop, there may be a concern that Z_i is endogenous since officers may choose their shifts, or that certain officers are assigned to certain shifts based on their willingness to search. While this may induce a correlation between Z_i and T_i *across* officers, the proposed methods remain valid as long as an *individual* officer's threshold is independent of Z_i since officers are evaluated separately. I assume this to be the case in the application. However, if individual officers prefer to search more during the night than during the day, and white and minority motorists drive at different times, then the test may conflate racial bias with the changes in thresholds stemming from Z_i.

Another challenge with this second approach to choosing an IV is that W_i must include all factors underlying the correlation between Z_i and T_i, and W_i must be observed. For instance, if W_i includes the fatigue of the officer, a variable that may vary by time and day but is never reported in the data, then it is impossible to condition on W_i. In such cases, the test may conflate changes in thresholds associated with W_i with racial bias.

3.2.3 Other applications

The proposed methods are not specific to police traffic searches and extend to other settings where an outcome test is appropriate. This may include parole release (Mechoulan and Sahuguet, 2015), pretrial detention (Arnold et al., 2018, 2022), mortgage approvals (Dobbie et al., 2021), and the labor market (Becker, 1957). More generally, the methodology non-parametrically identifies treatment effects on the thresholds in models similar to (1), with the treatment being the driver's race in my application. Treatments can be discrete, and the latent index compared against the threshold can be generalized to an expectation.

4 Detecting and measuring racial bias

In line with Becker (1957, 1993), my method tests whether an officer's search decisions are consistent with him being unbiased. If they are not, then the officer is deemed biased.

4.1 Defining the test

For each traffic stop, I observe the driver's race, R_i; the setting of the stop, Z_i; the search decision, $Search_i$; and whether contraband is found, Hit_i. By the law of iterated expectations and Assumption 1, the officer's search and hit rates for race $r \in \{w, m\}$ and setting $z \in \mathcal{Z}$ are[22]

$$\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} = \int_{\mathcal{V}} F_{T|R}(G(r,v,z) \mid r)\ dF_{V|R,Z}(v \mid r, z), \tag{2}$$

$$\mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} = \int_{\mathcal{V}} G(r,v,z)\ F_{T|R}(G(r,v,z) \mid r)\ dF_{V|R,Z}(v \mid r, z). \tag{3}$$

The conditional hit rate is the probability that contraband is found conditional on a traffic search and is equal to the ratio of (3) and (2),

$$\mathbb{P}\{Guilty_i = 1 \mid Search_i = 1, R_i = r, Z_i = z\} = \frac{\mathbb{P}\{\overbrace{Search_i \times Guilty_i}^{Hit_i} = 1 \mid R_i = r, Z_i = z\}}{\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\}}.$$

The instrument Z_i varies the search and hit rates by varying the distributions of risk.

Let $\mathcal{F}$ denote the space of distributions of $(V_i, T_i, Guilty_i, R_i, Z_i)$ satisfying Assumption 1. The sharp identified set of the model is[23]

$$\mathcal{F}^{\text{ID}} \equiv \{F \in \mathcal{F} : (2) \text{ and } (3) \text{ are satisfied for all } (r, z) \in \{w, m\} \times \mathcal{Z}\}.$$

To test for racial bias, the parameters of interest are only $F_{T|R}(\cdot \mid w)$ and $F_{T|R}(\cdot \mid m)$. So I consider a projection of the identified set.

To define this projection, let

$$G_i \equiv G(R_i, V_i, Z_i),$$

[22] See Online Appendix A.2 for the full derivation.

[23] The binary nature of outcomes $Search_i$ and $Guilt_i$ implies (2)–(3) contain all restrictions placed by the model on the observed data. Sharpness of $\mathcal{F}^{\text{ID}}$ immediately follows from its definition.

$$\sigma(\cdot;r) \equiv F_{T|R}(\cdot \mid r),$$

where G_i denotes the risk in stop i, and $\sigma(g;r)$ denotes the probability a driver with risk g and race r is searched. The function $\sigma(\cdot;r)$ represents the distribution of the officer's threshold for race r and is the parameter of interest. Denote the distribution of risk conditional on race and setting by

$$F_{G|R,Z}(g \mid r,z) \equiv \int_{\mathcal{V}} \mathbb{1}\{G(r,v,z) \leq g\} \; dF_{V|R,Z}(v \mid r,z).$$

Equations (2)–(3) may then be written as

$$\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} = \int_0^1 \sigma(g;r) \; dF_{G|R,Z}(g \mid r,z), \tag{4}$$

$$\mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} = \int_0^1 g \; \sigma(g;r) \; dF_{G|R,Z}(g \mid r,z). \tag{5}$$

Let $\mathcal{F}_T$ denote the space of non-decreasing, right-continuous functions with domain and codomain equal to $[0,1]$ (i.e., space of distributions of thresholds); and let $\mathcal{F}_G$ denote the space of distributions for scalar random variables with support $[0,1]$ (i.e., space of distributions of risk). Then the sharp identified set for the distribution of the officer's threshold is[24]

$$\Sigma \equiv \left\{ (\sigma(\cdot;w), \sigma(\cdot;m)) \in \mathcal{F}_T \times \mathcal{F}_T : \begin{array}{c} \exists F_{G|R,Z}(\cdot \mid r,z) \in \mathcal{F}_G \text{ s.t. (4) and (5) are} \\ \text{satisfied for all } (r,z) \in \{w,m\} \times \mathcal{Z} \end{array} \right\}. \tag{6}$$

A testable implication for racial bias immediately follows from (6).

Corollary 1. *Define* $\Sigma^{\text{Unb}} \equiv \{\sigma \in \mathcal{F}_T : (\sigma,\sigma) \in \Sigma\}$. *Under* (1) *and Assumption 1, if the officer is unbiased, then* Σ^{Unb} *is non-empty.*

Proof. Corollary 1 follows immediately from Definition 1. ■

[24]The sharpness of Σ follows from $\mathcal{F}^{\text{ID}}$ being sharp, and Σ being a projection of $\mathcal{F}^{\text{ID}}$.

A test built around Corollary 1 will be conservative since Σ^{Unb} may be non-empty even when the officer is biased. Nevertheless, since Σ is sharp, Corollary 1 is the strongest testable implication of the model for unbiasedness.

4.2 Intuition

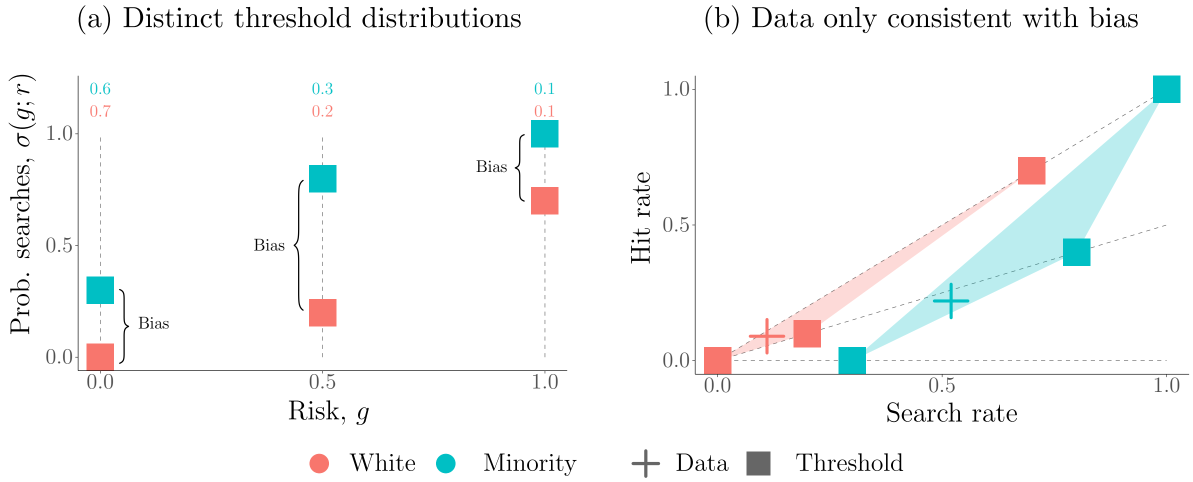
To build intuition for the test, consider a simple setting where risk is equal to 0, 0.5, or 1. The left panel of Figure 2 shows a distribution of thresholds, with each square indicating the probability that the officer searches a driver with a given level of risk. The right panel shows the data that can be generated by the random threshold. The horizontal position of each square in the right panel is equal to the search probability $\sigma(g;r)$ for some risk g; and the vertical position is equal to the joint probability of searching the driver and finding contraband, $g\sigma(g;r)$.

Equations (4)–(5) imply that the search and hit rates must lie in the convex hull of the three squares in the right panel, indicated by the purple region. Since the observed search and hit rates for both groups of drivers—represented by the crosses—indeed lie in the purple region, it is possible that both data points are generated by the same distribution of thresholds and it cannot be ruled out that the officer is unbiased. The colored numbers on the left panel indicate possible distributions of risk that generate crosses of the same color.

Figure 3 presents the case where only the red cross lies in the convex hull generated by the distribution in the left panel. This implies that the blue cross is generated by a different distribution. Corollary 1 states that if the officer is unbiased, then there must exist a distribution of thresholds that generates a purple region in the right panel containing both data points, as in Figure 2. If no such distribution exists, then the data for white and minority drivers must be generated by distinct distributions of thresholds and the officer must be biased.

Figure 1 presents the case where no single distribution of thresholds can generate the data for both groups of drivers. Racial bias is therefore detected. Note that bias is detected

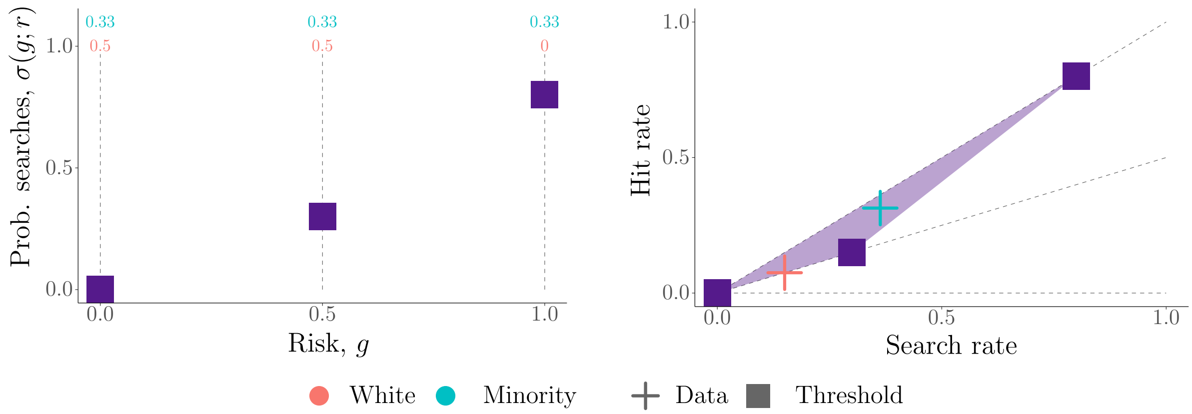
Figure 1: How search and hit rates are informative of bias



Note: Since it is impossible to find a single distribution of thresholds generating the data for both white and minority drivers, the officer must be biased. Any pair of thresholds required to generate the data for both groups of drivers implies an intensity of bias at each level of risk. By exploring the space of distributions of thresholds consistent with the data, I am able to derive bounds on various measures of bias (e.g., bias conditional on risk, bias averaged over risk).

Figure 2: How search and hit rates are generated

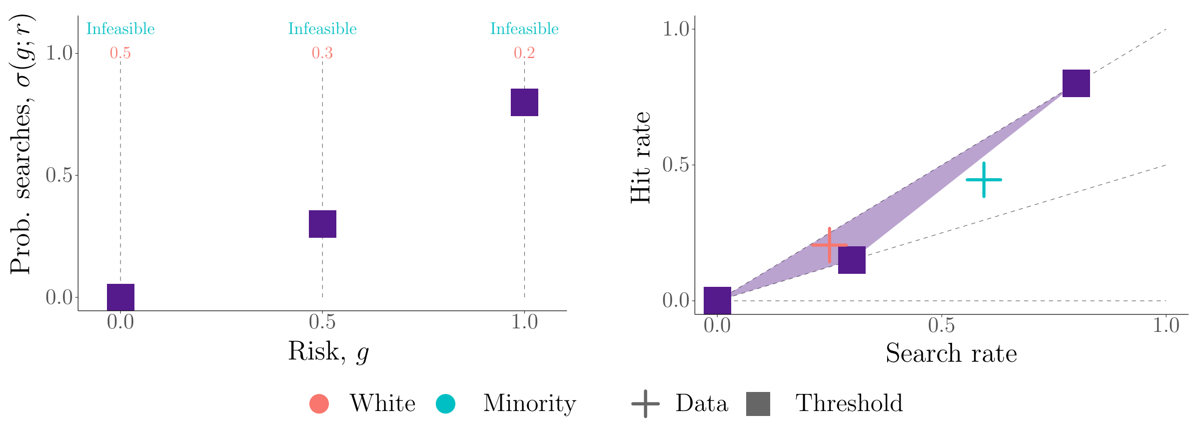
(a) Threshold distribution (b) Data consistent with threshold distribution



Note: Squares represent the officer's random threshold. Crosses represent observed search and hit rates. Colored numbers in the left panel indicate possible distributions of risk generating data points of the same color. Data consistent with the random threshold lie inside the purple region in the right panel. Since all data points lie inside the purple region, it is possible that they are generated by the distribution of thresholds in the left panel.

Figure 3: How search and hit rates are informative of thresholds

(a) Threshold distribution (b) Data inconsistent with threshold distribution



Note: The red data point is consistent with the random threshold shown but the blue data point is not. If there exists a different distribution of thresholds consistent with both data points, the officer may be unbiased. Otherwise, the officer must have distinct distributions for white and minority drivers.

even though there is only one data point for each race of drivers, which corresponds to the case of having no instrument or an irrelevant instrument.[25] If an instrument were available and relevant, there would be multiple data points for each race of drivers. This strengthens the test by making it more difficult to find a single distribution of thresholds capable of generating all the data points.

My framework further allows me to obtain bounds on the intensity of bias. The left panel of Figure 1 shows two distinct distributions of thresholds that could have generated the data in the right panel, as well as the implied intensity of bias at each level of risk. By considering different distributions of thresholds and risks that are consistent with the data, I obtain bounds on various measures of bias.

4.3 Implementation

Corollary 1 may be implemented as a bilinear programming (BP) problem. Despite being non-convex, bilinear programs can be solved to provable global optimality by commercial solvers.[26] For simplicity, suppose that G_i is discrete and $\operatorname{supp}(G_i) = \{g_1, \ldots, g_K\}$ for finite K. Then (4)–(5) become

$$\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} = \sum_{k=1}^{K} \underbrace{\sigma(g_k; r)\ p_{r,z}(g_k)}_{\text{Bilinear terms}}, \tag{7}$$

$$\mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} = \sum_{k=1}^{K} g_k \underbrace{\sigma(g_k; r)\ p_{r,z}(g_k)}_{\text{Bilinear terms}}, \tag{8}$$

[25]This can also occur when the instrument is relevant, but Z_i only correlates with components of V_i that are uninformative of risk.

[26]Bilinear programs are solved to global optimality using a branch-and-bound algorithm. The domain space is partitioned, and convex McCormick relaxations of the original problem are solved across the partitions. See McCormick (1976), Mehlhorn et al. (2008), and Gurobi Optimization, Inc. (2021).

where

$$p_{r,z}(g) \equiv \mathbb{P}\{G_i = g \mid R_i = r, Z_i = z\}$$

denotes the distribution of risk conditional on the race of the driver and setting of the stop. Online Appendix D discusses how B-splines may be used to model the distributions of thresholds and risk if G_i is continuously distributed.

To specify the BP problem, I introduce the following notation.

$$\begin{aligned}
\mathbf{a}^S_{r,z} &\equiv \mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} \\
\mathbf{a}^H_{r,z} &\equiv \mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} \\
\mathbf{g} &\equiv (g_1, \ldots, g_K)' \\
\boldsymbol{\sigma}_r &\equiv (\sigma(g_1; r), \ldots, \sigma(g_K; r))' \\
\mathbf{p}_{r,z} &\equiv (p_{r,z}(g_1), \ldots, p_{r,z}(g_K))'
\end{aligned}$$

The probabilities $\mathbf{a}^S_{r,z}$, $\mathbf{a}^H_{r,z}$ are the search and hit rates for each race r and setting z and are identified from the data. The vector $\mathbf{g}$ is the support of G_i, which I assume is known to the researcher. The unknown parameters of the BP problem are $\{\boldsymbol{\sigma}_r\}_{r\in\{w,m\}}$, which are the values of $\sigma(\cdot; r) \in \mathcal{F}_T$ evaluated at each point of $\mathbf{g}$; and $\{\mathbf{p}_{r,z}\}_{(r,z)\in\{w,m\}\times\mathcal{Z}}$, which are the distributions of risk conditional on race and setting. For brevity, I refer to the distributions of thresholds by $\{\boldsymbol{\sigma}_r\}$ and the distributions of risk by $\{\mathbf{p}_{r,z}\}$.

To ensure these parameters are consistent with the model, I impose two baseline sets of constraints, both of which are linear in the parameters of the model. The first set of constraints is

$$0 \leq \boldsymbol{\sigma}_{r,k} \leq \boldsymbol{\sigma}_{r,k+1} \leq 1 \text{ for } r \in \{w, m\} \text{ and } k = 1, \ldots, K-1, \tag{9}$$

where $\boldsymbol{\sigma}_{r,k}$ denotes the k^{th} component of $\boldsymbol{\sigma}_r$, i.e., $\boldsymbol{\sigma}_{r,k} = \sigma(g_k; r)$. This ensures $\sigma(\cdot; r)$ is a valid CDF.[27] The second set of constraints is

$$\mathbf{p}_{r,z,k} \in [0,1] \text{ for } (r,z) \in \{w,m\} \times \mathcal{Z} \text{ and } k = 1, \ldots, K, \tag{10}$$

$$\sum_{k=1}^{K} \mathbf{p}_{r,z,k} = 1 \text{ for } (r,z) \in \{w,m\} \times \mathcal{Z}, \tag{11}$$

where $\mathbf{p}_{r,z,k}$ denotes the k^{th} component of $\mathbf{p}_{r,z}$. This ensures $\mathbf{p}_{r,z}$ generates a valid distribution of risk. To simplify the discussion, I assume that $\operatorname{supp}(Z_i \mid R_i = w) = \operatorname{supp}(Z_i \mid R_i = m)$, but this assumption is not necessary.

Define the population criterion function as

$$Q(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \equiv \sum_{r,z} \Big| \overbrace{\boldsymbol{\sigma}_r' \mathbf{p}_{r,z}}^{\text{Bilinear terms}} - \mathbf{a}_{r,z}^S \Big| + \sum_{r,z} \Big| \overbrace{(\mathbf{g} \odot \boldsymbol{\sigma}_r)' \mathbf{p}_{r,z}}^{\text{Bilinear terms}} - \mathbf{a}_{r,z}^H \Big|,$$

where $\odot$ denotes the Hadamard (element-wise) product. The criterion function measures how much (7)–(8) are violated. The bilinear terms account for endogeneity in search decisions by linking the observed search and hit rates to the selection mechanism characterized by $\{\boldsymbol{\sigma}_r\}$, which allows the test to avoid the inframarginality problem.

The following proposition describes how to test for bias in population using Corollary 1.

Proposition 1. *Define $Q^\star$ as*

$$Q^\star \equiv \min_{\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}} Q(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \tag{12}$$

$$\text{s.t.} \quad \boldsymbol{\sigma}_w = \boldsymbol{\sigma}_m, \ (9), \ (10), \ (11).$$

The officer is biased if $Q^\star > 0$.

[27] Constraint (9) differs from the standard monotonicity restriction invoked in the judge IV setting, which imposes a nesting condition that induces a ranking of leniency across decision makers (Imbens and Angrist, 1994; Chyn et al., 2025). Constraint (9) ensures that the decisions of a single officer is consistent with the model. I impose no restrictions on how one officer's decisions compare against that of another.

Proof. The constraint $\boldsymbol{\sigma}_w = \boldsymbol{\sigma}_m$ restricts the officer to be unbiased. If $Q^\star > 0$, then (7) or (8) is violated for some $(r, z) \in \{w, m\} \times \mathcal{Z}$. Then by Corollary 1, the officer is biased. ■

The criterion $Q^\star$ in Proposition 1 is the minimum ℓ^1-norm between the moments of the model and the moments of the data when the officer is restricted to be unbiased. Since the ℓ^1-norm can be reformulated as being linear, the criterion function in (12) is bilinear. Other norms may be used but may be more computationally demanding.

4.3.1 Adding restrictions

The test can be strengthened by restricting $\mathcal{F}_T$ and $\mathcal{F}_G$. This is straightforward to do if the restrictions can be written as linear, bilinear, or quadratic constraints to the BP problem.

For example, consider restricting the mass of drivers to be decreasing as risk increases,

$$\mathbf{p}_{r,z,k} \geq \mathbf{p}_{r,z,k+1} \text{ for } (r, z) \in \{w, m\} \times \mathcal{Z} \text{ and } k = 1, \ldots K - 1. \tag{13}$$

Such an assumption is suitable when the mass of low-risk drivers in population is large compared to the mass of high-risk drivers. Even if the officer is much more likely to stop high-risk drivers, the greater volume of low-risk drivers on the road may result in a distribution of risk (conditional on being stopped) where the mass of drivers decreases as risk increases.[28]

Figure 4 demonstrates how this restriction strengthens the test. The same distribution of thresholds is depicted in the left and right panel. However, the range of data that can be generated by the threshold is reduced when (13) is imposed. In fact, while there exist random thresholds capable of generating the data for both races when there are no restrictions on the distributions of risk, it is no longer the case once (13) is imposed.

For more examples of imposing restrictions on the model, see Online Appendix C.

[28] See Online Appendix C.3 for a numerical example.

Figure 4: Strengthening the test by restricting $\{\mathbf{p}_{r,z}\}$

(a) Feasible region without restriction (13) (b) Feasible region with restriction (13)

Note: The purple region in the left panel shows the possible data points generated by a particular random threshold when there are no restrictions on the distribution of risk. The purple region in the right panel shows the possible data points generated by the same random threshold, except the mass of drivers is restricted to be decreasing as risk increases. Reducing the size of the purple region strengthens the test for racial bias by making it easier to rule out distributions from Σ^{Unb}.

4.4 Determining the direction and intensity of bias

If bias is detected, the next step is to determine how the officer is biased. This can be done in several ways. Below, I first introduce a general measure of bias and show how it can be bounded. I then show some restrictions that can be imposed to obtain specific measures of bias.

4.4.1 Bounding a general measure of bias

The general measure of bias takes the form

$$\theta \equiv \boldsymbol{\omega}' \underbrace{(\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w)}_{\beta(\cdot)}, \tag{14}$$

where $\boldsymbol{\omega} = (\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_K)'$ is a vector of weights with $\boldsymbol{\omega}_k \in [0,1]$ for $k = 1, \ldots, K$ and $\sum_{k=1}^{K} \boldsymbol{\omega}_k = 1$. θ is thus a weighted average of the intensity of bias at each level of risk and $\boldsymbol{\omega}$ is a counterfactual distribution of risk. The choice of $\boldsymbol{\omega}$ determines the measure of bias, and the weights can be chosen beforehand or treated as variables in the BP problem. If $\theta > 0$ ($\theta < 0$), then the officer is biased against minorities (whites) given $\boldsymbol{\omega}$.

Proposition 2. *The sharp bounds on θ are obtained by solving the following BP problem,*

$$\theta_{\text{lb}}, \theta_{\text{ub}} \equiv \min_{\boldsymbol{\omega}, \{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}}/\max \; \boldsymbol{\omega}' (\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w) \tag{15}$$

$$\text{s.t.} \quad Q(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) = 0, \; (9), \; (10), \; (11).$$

Proof. The objective in (15) defines the measure of bias, θ. Since the constraints characterize the sharp identified set Σ, the bounds on θ are sharp by construction.[29] ■

Absent restrictions on $\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w$, the officer can be biased against one group of drivers for a given level of risk and reverse their direction of bias at another level of risk. If the researcher has a prior on the direction of bias, then a sign restriction on the elements of $\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w$ can easily be imposed. For example, bias against white drivers is ruled out if every element of $\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w$ is restricted to be non-negative.

4.4.2 Bounding bias conditional on risk

Bounds on bias conditional on risk are obtained by setting $\boldsymbol{\omega} = \mathbf{e}_k$, where $\mathbf{e}_k \in \mathbb{R}^K$ is the k^{th} standard basis vector. This choice of $\boldsymbol{\omega}$ implies

$$\theta = \sigma(g_k; m) - \sigma(g_k; w) = \beta(g_k).$$

[29] Let Θ denote the identified set for θ. The bounds in Proposition 2 are sharp in the sense that they are the smallest and largest values of Θ. However, because bilinear programs are non-convex, Θ need not be the full interval $[\theta_{\text{lb}}, \theta_{\text{ub}}]$. See Online Appendix B for how to fully recover Θ.

The researcher can therefore bound the bias at every level of risk. It is possible for $0 \in [\theta_{\text{lb}}, \theta_{\text{ub}}]$ for every level of risk even if the officer fails the test in Proposition 1. This corresponds to the case where bias is detected, but the direction of bias is undetermined.

4.4.3 Bounding average bias

Bounds on the average bias under a counterfactual distribution of risk can be obtained by setting $\boldsymbol{\omega}$ equal to that distribution. For example, the average bias when risk is uniformly distributed for either group of drivers corresponds to $\boldsymbol{\omega}_k = \frac{1}{K}$ for all $k = 1, \ldots K$.

Bounds on the average bias under the actual unobserved distribution of risk for white or minority drivers are also obtainable. For example, the following constraint sets $\boldsymbol{\omega}$ equal to the distribution of risk averaged across settings for white drivers,

$$\begin{aligned}
\boldsymbol{\omega}_k &= \mathbb{P}\{G_i = g_k \mid R_i = w\} \\
&= \sum_{z \in \mathcal{Z}} \mathbb{P}\{G_i = g_k \mid R_i = w, Z_i = z\}\, \mathbb{P}\{Z_i = z \mid R_i = w\} \\
&= \sum_{z \in \mathcal{Z}} \mathbf{p}_{w,z,k}\, \mathbb{P}\{Z_i = z \mid R_i = w\}, \qquad (16)
\end{aligned}$$

where the second equality follows by the law of iterated expectations, and $\mathbb{P}\{Z_i = z \mid R_i = w\}$ is identified from the data. This choice of $\boldsymbol{\omega}$ implies that $\theta = \mathbb{E}[\beta(G_i) \mid R_i = w]$, i.e, the change in search rates for white drivers if they were treated as minorities.

4.5 Estimation and inference

For partially identified models defined by moment (in)equalities, Bugni et al. (2015) propose a specification test, and Bugni et al. (2017) construct confidence intervals for subvectors of parameters. I adapt these methods to test for bias and construct confidence intervals on measures of bias.

4.5.1 Testing for bias

Let $\mathbf{a} \equiv (\mathbf{a}^S_{r,z}, \mathbf{a}^H_{r,z})'_{(r,z)\in\{w,m\}\times\mathcal{Z}}$ denote the vector of search and hit rates for all races and settings. Let $\mathbf{D}$ denote a diagonal matrix containing $Var[Search_i \mid R_i = r, Z_i = z]$ and $Var[Hit_i \mid R_i = r, Z_i = z]$ for all races and settings. Let $\widehat{\mathbf{a}}$ and $\widehat{\mathbf{D}}$ denote consistent estimates of $\mathbf{a}$ and $\mathbf{D}$. Let $\phi(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\})$ denote the vector of search and hit rates implied by the model parameters, i.e., the right hand sides of (7)–(8). The scaled sample criterion is

$$\widehat{Q}(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \equiv \sqrt{n} \left\| \widehat{\mathbf{D}}^{-1/2} \left(\phi(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) - \widehat{\mathbf{a}}\right) \right\|_1,$$

where n is the total number of traffic stops and $\|\cdot\|_1$ denotes the ℓ^1-norm.[30]

Define

$$\begin{aligned} \widehat{Q}^\star_{\text{Unb}} \equiv & \min_{\{\boldsymbol{\sigma}_r\},\{\mathbf{p}_{r,z}\}} \widehat{Q}(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \\ \text{s.t.} \quad & \boldsymbol{\sigma}_w = \boldsymbol{\sigma}_m, \ (9), (10), (11), \end{aligned}$$

and

$$\begin{aligned} \widehat{Q}^\star_{\text{Biased}} \equiv & \min_{\{\boldsymbol{\sigma}_r\},\{\mathbf{p}_{r,z}\}} \widehat{Q}(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \\ \text{s.t.} \quad & (9), (10), (11). \end{aligned}$$

I test the hypotheses

$$H_0 : Q^\star = 0 \quad \text{versus} \quad H_1 : Q^\star > 0,$$

where the null hypothesis corresponds to the officer being unbiased (see Proposition 1). The

[30] $\widehat{Q}$ is based on the scaled sample criterion proposed by Bugni et al. (2015) and the the Modified Method of Moments test function from Andrews and Guggenberger (2009). I use the ℓ^1-norm instead of the squared Euclidean norm.

test statistic

$$\widehat{\tau} \equiv \widehat{Q}^{\star}_{\text{Unb}} - \widehat{Q}^{\star}_{\text{Biased}} \tag{17}$$

compares the fit of the model when the officer is restricted to be unbiased against the fit without the restriction. A large test statistic suggests the fit of the model is affected by the restriction of unbiasedness and is evidence against the null hypothesis.

I estimate the distribution of $\widehat{\tau}$ under the null hypothesis by resampling the data B times. The data are resampled at the weekly level to account for dependencies over time. For each resampled dataset, indexed by $b = 1, \dots, B$, I calculate (17) and denote its value by $\widehat{\tau}_b$. I reject the null hypothesis at the α significance level if $\widehat{\tau}$ exceeds the $1-\alpha$ quantile of $\{\widehat{\tau}_b - \widehat{\tau}\}$.

4.5.2 Estimating the intensity of bias

When bias is detected, the bounds on bias are estimated by solving

$$\begin{aligned} \theta_{\text{lb}}, \theta_{\text{ub}} \equiv & \min_{\boldsymbol{\omega}, \{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}} / \max \; \boldsymbol{\omega}' \left(\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w\right) \\ \text{s.t.} \quad & \widehat{Q}(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \leq \widehat{Q}^{\star}_{\text{Biased}}, \; (9), (10), (11). \end{aligned}$$

Confidence intervals on the bounds are constructed by test inversion. That is, I test the specification that the intensity of bias is equal to $t \in [-1, 1]$. If the test does not reject the specification at the α significance level, then t enters the $(1-\alpha)$ confidence interval.[31]

5 Application

I apply the test to police traffic data from the Metropolitan Nashville Police Department (MNPD). The data contain records of traffic stops for over 2,200 MNPD officers between

[31]See Online Appendix B for a full description of this procedure.

2010 and 2019 and is made available by the Stanford Open Policing Project (Pierson et al., 2020).

5.1 Data

Each observation in the data is a traffic stop made by an officer. Observed driver characteristics include race, age, sex, and state of registration. Observed logistical details include the time and geocoordinates of the stop; the reason for the stop; whether a search occurred, and if so, why the search occurred and whether contraband was found. I categorize the reason for the stop into three groups: driving-related reasons, non-driving reasons, and investigative reasons.[32] Reasons for searches include driver consent, probable cause, and plain view of contraband. Although the data categorize contraband into weapons and drugs, I treat all forms of contraband as being the same.

I supplement the traffic data with MNPD data on criminal incidents and calls for services,[33] and local measures of racial composition and median household income from the American Community Survey (ACS). Both the MNPD and ACS data are at the census tract level and allow me to control for environmental factors that potentially correlate with the setting of the stop and the officer's thresholds.

5.2 Restricting the sample

To study bias in traffic searches, the searches analyzed must be discretionary. Mandatory searches such as inventory searches, warrant-based searches, and searches incidental to an

[32] Driving-related reasons include moving traffic violations, safety violations, and vehicle equipment violations. Non-driving reasons include seat belt violations, parking violations, registration violations, and issues with child restraints. Investigative stops are its own category.

[33] I use criminal incidents and calls for services related to violent crimes, theft, or drugs, as these may affect an officer's decision to search for contraband.

Table 1: Summary of stops, searches, and hits for select 50 officers

	Full sample		Avg. by officer	
	White	Minority	White	Minority
Stops	109,023	113,405	2,180	2,268
Searches	12,622	15,732	252	315
Hits	1,831	2,741	37	55
Search rate	0.1158	0.1387	0.1546	0.1884
Uncon. hit rate	0.0168	0.0242	0.0277	0.0297
Con. hit rate	0.1451	0.1742	0.2431	0.2135

Notes: For each officer, the conditional hit rate can be calculated from the ratio of the unconditional hit rate and search rate.

arrest are thus excluded.[34] In total, 72% of the traffic searches in the data are retained.

Since officers are evaluated separately, I evaluate the 50 officers with the most searches to obtain reliable estimates of their search and hit rates. On average, each officer has made 2,180 stops and 252 searches for white drivers, and 2,268 stops and 315 searches for minority drivers. Surprisingly, one third of all searches in the data are conducted by these officers.

Finally, I restrict the data to white, black, and Hispanic drivers, and define "Minority" drivers to be either black or Hispanic.

Table 1 summarizes the number of traffic stops, searches, and hits in the final sample.

5.3 Control and instrumental variables

Z_i is combinations of the day of the week and the patrol shift. Days are divided into weekdays and weekends, and patrol shifts are either in the morning (7 a.m. to 3 p.m.), evening (3 p.m. to 11 p.m.), or night (11 p.m. to 7 a.m.). This generates up to six values of Z_i for each officer. To support Assumption 1(ii), I control for variables that may correlate with both Z_i

[34] Inventory searches are required when a vehicle is impounded. Warrants are typically obtained before the traffic stop, suggesting that warrant-based searches are predetermined and non-discretionary. Searches incidental to an arrest occur after a driver has been arrested.

Table 2: Summary of control variables

	Drivers stopped		Drivers searched	
	White	Minority	White	Minority
Driver characteristics				
Male	0.6032	0.6007	0.6613	0.7722
Age	37.28	34.64	32.31	30.49
Out of state	0.0638	0.0330	0.0490	0.0340
Reason for stop				
Driving	0.8803	0.8776	0.8668	0.8687
Non-driving	0.1070	0.1065	0.1072	0.1031
Investigation	0.0127	0.0159	0.0260	0.0282
Reason for search				
Plain view			0.4978	0.2606
Consent			0.4336	0.5938
Probable Cause			0.0686	0.1456
Location				
Highway	0.1228	0.0644	0.0759	0.0495
Precinct 1	0.0763	0.0509	0.0640	0.0521
Precinct 2	0.1190	0.1760	0.0882	0.1920
Precinct 3	0.1042	0.1446	0.0913	0.1377
Precinct 4	0.0395	0.0249	0.0789	0.0381
Precinct 5	0.3618	0.2567	0.2573	0.2227
Precinct 6	0.0400	0.1100	0.0257	0.0774
Precinct 7	0.1366	0.1528	0.1469	0.1540
Precinct 8	0.1225	0.0842	0.2477	0.1260
Census tract demographics				
Percent white	0.5901	0.4523	0.6028	0.4580
Median household income	49038	41170	48642	40029
Crime incident rate	0.0256	0.0369	0.0305	0.0400
Calls for MNPD services	0.0207	0.0216	0.0212	0.0227

Notes: Crime and call rates are per capita and are restricted to those pertaining to violent crimes, theft, or drugs.

and officer thresholds. Table 2 summarizes the distribution of control variables.

The first set of controls consists of observable characteristics of the driver besides race, which includes age, sex, and state of registration. This set of controls accounts for how officers may feel differently towards searching certain demographics who may drive during different times of the day and days of the week.

The second set of controls include the details of the traffic encounter, namely the reason for the stop and, if a search took place, the reason for the search.[35] These variables control for how certain aspects of the traffic stop (e.g., being stopped for driving-related reasons) or driver behavior (e.g., having contraband in plain view) might affect an officer's thresholds and be correlated with the setting. For example, Makofske (2020) finds that officers in Louisville, Kentucky arrest 40% of drivers stopped for failing to signal, compared to 1% of drivers stopped for any other reason. This suggests that certain stops in Louisville are pretextual and the reason for stopping a driver can affect an officer's thresholds. Although the MNPD data do not show signs of pretextual stops, they show a 10% increase in the proportion of stops being attributed to driving-related reasons across the evening and night shifts,[36] as well as a 50% increase in searches attributed to contraband being in plain view across the same pair of shifts. Controlling for these features of the traffic stop reduces the concern that the test is detecting differences along these dimensions rather than detecting racial bias.

The final set of controls relates to the environment where the stop takes place. This includes whether the stop was made on a street or a highway; which police precinct the stop was made in; the racial composition, household income, and crime rate of the census tract; and the frequency of calls for MNPD services originating from the census tract. This accounts for the possible correlation between an officer's thresholds and Z_i induced by his

[35] Durlauf and Heckman (2020) raise concerns about the credibility of self-reported police data. While the concern is valid, there is currently not a good solution.

[36] In a study on endogenous driving behavior, Kalinowski et al. (2023) find that minority drivers adjust their driving behavior during the day, when their race is more visible to the officer.

surroundings. As discussed in Section 3.2.2, such a correlation may arise if officers are more likely to be in dangerous or high-crime neighborhoods at certain times (e.g., night shifts on weekends), and officers are more concerned about their safety or face higher opportunity costs to searching vehicles when in these neighborhoods.[37]

Some potential concerns with the data are endogenous shift assignments, ticket quotas, or officers being instructed to search more aggressively during certain times. Regarding endogenous shift assignments, see the discussion in Section 3.2.2. Regarding the ticket quotas, Tennessee has explicit laws banning quotas on traffic citations. Although this has not stopped departments from implementing such quotas, ticket quotas pertain to stop decisions of officers instead of search decisions.[38] As long as ticket quotas do not affect search thresholds, then the quotas only impact the search decisions by affecting the distribution of risk via sample selection. Regarding the concern that officers are instructed to search more aggressively during different shifts, there were no such policies during the time frame of the data I analyze. To the best of my knowledge, such policies were only implemented beginning in July of 2019.[39]

[37] Roh and Robinson (2009) find there to be spatial correlation in traffic search decisions even after controlling for driver characteristics. The authors attribute the correlation to similarities in environmental variables, such as the racial composition of the neighborhood and the volume of police allocated nearby. Novak and Chamlin (2012) also find that the police workload (measured via calls for services) and degree of 'social disorganization' (e.g., percentage of single parent households, percentage of residents in poverty) are predictive of officer behavior.

[38] The mayor of Ridgetop, TN attempted to have the city's police department enforce a ticket quota to raise money for the city, only to be turned in by the city's police chief (Ferrier, 2019). See Tennessee Code §39-16-516 (2014) for the law banning ticket quotas.

[39] In July of 2019, the MNPD introduced the Entertainment District Initiative, which assigned 17 additional officers to the Entertainment District on Fridays and Saturdays between 6 p.m. and 4 a.m. to improve public safety. These officers performed high-visibility patrols on foot, bike, and utility task vehicles, and would make unannounced visits to local establishments. In February of 2021, the MNPD introduced the Office of Alternative Policing Strategies to address an increase in violent crime in Nashville. A new shift of 80 officers working between 5:30 p.m. and 3:30 a.m. was added across all precincts to perform high-visibility

A limitation of the data is the absence of criminal records for drivers. Controlling for this information is important if officer thresholds are believed to depend on past offenses of drivers. Unfortunately, police traffic stop data do not contain such information. Identifying information of drivers is also typically hidden, making it impossible to merge in criminal records for drivers. The data set constructed by Feigenberg and Miller (2022) is unique in its inclusion of driver criminal histories.

5.4 Setting up the BP problem

I discretize the support of risk to be

$$\mathbf{g} = \{\underbrace{0, 0.025, 0.05, 0.075}_{\text{Increments of } 0.025}, \underbrace{0.1, 0.15, 0.20, 0.25}_{\text{Increments of } 0.05}, \underbrace{0.3, 0.4, 0.5, 0.6}_{\text{Increments of } 0.1}, 0.75, 1\}.$$

I choose $\mathbf{g}$ to be finer at lower levels of risk since Table 1 shows that the average conditional hit rates are between 21% and 24%, which suggests most drivers searched are relatively low-risk. Table 3 presents the conditional hit rates for each setting after accounting for controls. The average conditional hit rates remain low, ranging from 5% to 27%. The model also implies that drivers who are searched represent the riskiest subset of drivers who are stopped. In conjunction with the low conditional hit rates, this further suggests that most drivers stopped are low-risk. I incorporate this into the model by imposing the monotonicity restriction in (13), requiring that $\mathbf{p}_{r,z}$ be decreasing as risk increases for all $(r, z) \in \{w, m\} \times \mathcal{Z}$.

I do not impose any restrictions on $\boldsymbol{\sigma}$ except that it is non-decreasing in risk (as implied by Assumption 1) and lies in the unit interval.

The probabilities $\widehat{\mathbf{a}}^S_{r,z}$, $\widehat{\mathbf{a}}^H_{r,z}$ are estimated using logistic regressions. Since traffic searches and hits can be rare events for officers, I use Firth's logistic regression with intercept-correction to obtain unbiased estimates of the search and hit rates (Puhr et al., 2017).[40]

patrols to deter and detect violent crimes. See Aaron et al. (2019), Rau (2021), and McDonald (2021).

[40] Puhr et al. (2017) show that Firth's logistic regression debiases predicted probabilities for rare events

Table 3: Search and conditional hit rates by Z_i

		White		Minority	
Day	Shift	Search	Cond. Hit	Search	Cond. Hit
Weekday	Morning	0.0376	0.2617	0.0603	0.2265
Weekday	Evening	0.1268	0.1774	0.1528	0.1826
Weekday	Night	0.2711	0.1080	0.2381	0.1645
Weekend	Morning	0.0372	0.2656	0.1091	0.1272
Weekend	Evening	0.1349	0.1044	0.1259	0.1597
Weekend	Night	0.2753	0.0562	0.2334	0.1064
Mean		0.1158	0.1958	0.1387	0.1868

Notes: Search and conditional hit rates account for the control variables. The mean rates for the observed data are calculated by weighting each setting by the proportion of stops in the data made in each setting, and taking a weighted average of the rates across the settings.

To construct $\widehat{\mathbf{a}}^S_{r,z}$, I first regress Search_i on setting Z_i and controls X_i conditional on race $R_i = r$. This provides an estimate of $\mathbb{P}\{\mathit{Search}_i = 1 \mid R_i = r, Z_i = z, X_i = x\}$. I then set $\widehat{\mathbf{a}}^S_{r,z}$ equal to the predicted probabilities averaged over the sample distribution of X_i for either race of drivers, i.e.,

$$\widehat{\mathbf{a}}^S_{r,z} = \widehat{\mathbb{E}}\left[\widehat{\mathbb{P}}\{\mathit{Search}_i \mid R_i = r, Z_i = z, X_i\} \,\middle|\, R_i = r'\right] \text{ for } r' \in \{w, m\}. \tag{18}$$

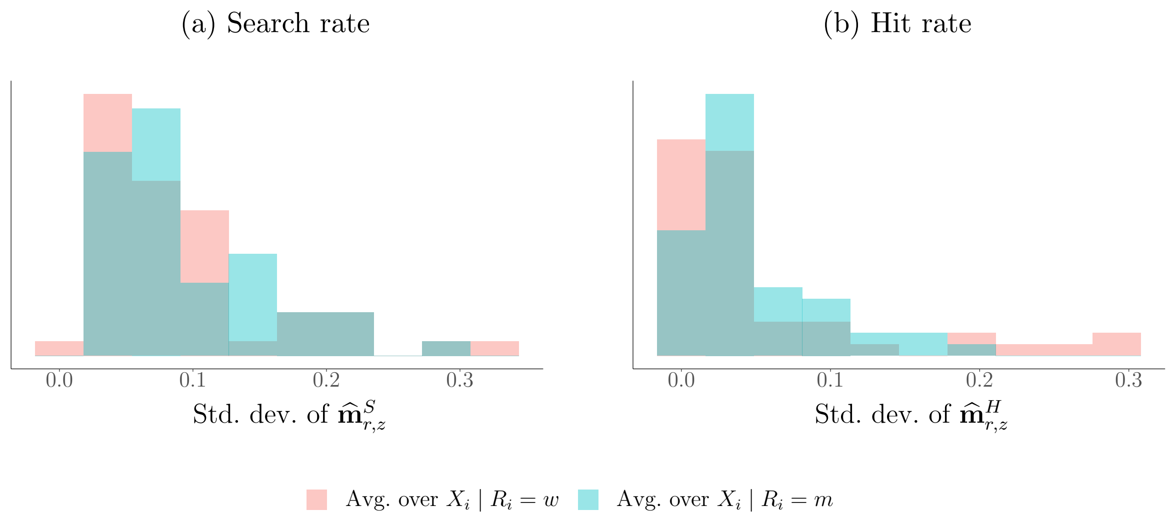
This allows me to control for X_i such that the estimates are representative of each race. In Section 5.5, I present results for both $r' = w$ and $r' = m$.

The hit rates $\widehat{\mathbf{a}}^H_{r,z}$ are estimated as in (18), except I regress Hit_i on Z_i and X_i conditional on each race.

Figure 5 summarizes the variation in search and hit rates generated by Z_i within officer. The figure is obtained by calculating the standard deviations of $\{\widehat{\mathbf{a}}^S_{r,z}\}$ and $\{\widehat{\mathbf{a}}^H_{r,z}\}$ across R_i and Z_i for each officer, and then presenting the histogram of these standard deviations. Greater variation in search and hit rates increases the power of the test by making it more

and outperforms competing methods, including King and Zeng (2001).

Figure 5: Variation in search and hit rates across Z_i



Note: The left (right) panel shows the distribution of the standard deviation of search (hit) rates across R_i and Z_i. The standard deviation of the search (hit) rates across R_i and Z_i is calculated for each officer, and the histograms show the distribution of those standard deviations. The histograms in red (blue) correspond to the case where $\widehat{\mathbf{a}}^S_{r,z}$ and $\widehat{\mathbf{a}}^H_{r,z}$ are obtained by averaging the fitted search and hit rates from logistic regressions over the distribution of $X_i \mid R_i = w$ ($X_i \mid R_i = m$).

difficult to find a single distribution of thresholds generating the data for both groups of drivers.[41]

5.5 Results

When averaging the search and hit rates over $X_i \mid R_i = w$, I reject the null hypothesis that the officer is unbiased at the 5% significance level for four of the 50 officers. In addition, three officers are at the margin of failing the test, i.e., the null hypothesis may only be rejected for them at the 6% significance level.

When averaging the search and hit rates over $X_i \mid R_i = m$, I again reject the null

[41]There are three officers with no variation in hit rates as they have never found contraband despite having searched many drivers. For these officers, the criterion and test exclusively depend on (7).

hypothesis for four officers. Compared to the previous case where search and hit rates were averaged over $X_i \mid R_i = w$, two of these officers also failed the test, and one of these officers was at the margin of failing.

Correcting for multiple hypothesis testing using the Holm-Bonferroni method, I reject the null hypothesis for two officers whether averaging search and hit rates over $X_i \mid R_i = w$ or $X_i \mid R_i = m$. Given the conservative natures of the test and Holm-Bonferroni method, I focus on the estimates obtained without correcting for multiple hypothesis testing.

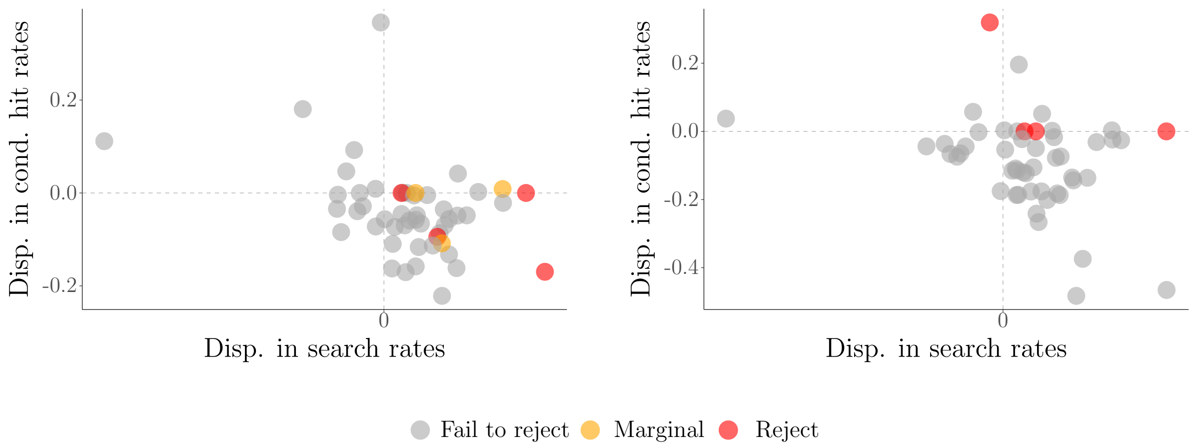
The results suggest that bias depends on observable characteristics of the driver and traffic stop, X_i. Table 2 compares the distribution of X_i for white and minority drivers, and shows that minority drivers are on average younger; are stopped in different precincts; and are stopped in areas with higher crime rates, lower income, and lower proportion of white residents. However, these differences in X_i are balanced across both groups of drivers when testing for bias, so the test is not conflating differences in X_i across race with differences in thresholds across race.

To see whether the proposed methods imply a simpler approach to detecting bias, Figure 6 shows the racial disparities in search and conditional hit rates for officers who are flagged as racially biased and those who are not. Positive disparities in search (hit) rates indicate that minority drivers have higher search (hit) rates compared to whites. The left panel averages search and hit rates over the distribution of $X_i \mid R_i = w$, and the right panel averages the rates over the distribution of $X_i \mid R_i = m$. There is no clear relationship between search and hit rate disparities and racial bias.

Figure 7 presents the data for an officer for whom bias is not detected. The circles in the left panel represent the search and hit rates by race and setting, and the size of the circles indicate the number of stops associated with the setting. The purple region shows the set of data points that are consistent with the threshold distribution in the right panel. Since all the data points lie inside the purple region, it is possible for the observed data for white and minority drivers to be generated by the same random threshold. Applying the test from

Figure 6: Racial disparities in search and conditional hit rates by officer

(a) Rates averaged over $X_i \mid R_i = w$ (b) Rates averaged over $X_i \mid R_i = m$

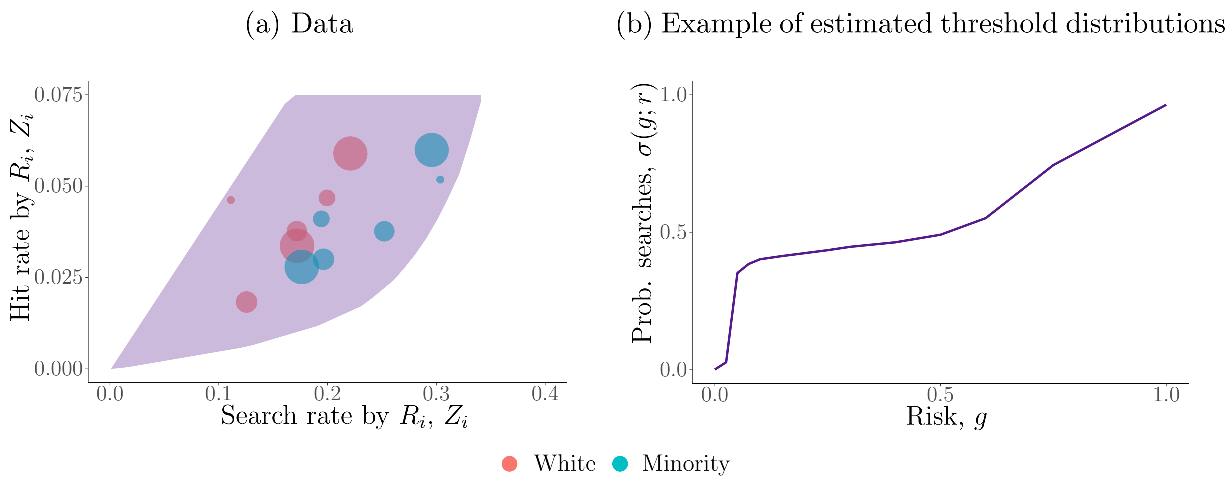


Note: Each point corresponds to an individual officer. Search and hit rates of each officer are averaged across the different settings, controlling for observed characteristics of the driver. Positive disparities indicate that minority drivers have higher rates compared to white drivers. Red points indicate officers for whom the null hypothesis of being unbiased is rejected at the 5% significance level. Orange points indicate officers for whom the null hypothesis is close to being rejected (rejection at the 6% significance level). Grey points indicate the remaining officers for whom the null hypothesis is not rejected.

Section 4, I cannot reject the null hypothesis that the officer is unbiased at the 5% significant level.

Figure 8 presents the data for an officer for whom bias is detected. The top right panel presents one possible estimate of the officer's distribution of thresholds. The bottom panel presents the estimated bounds on the bias, with the gray band showing the bounds conditional on risk, and the dashed lines showing the bounds on the average bias. The red (blue) dashed lines indicate the average bias when the distribution of risk is consistent with that of white (minority) drivers in the data. The bounds on the bias conditional on risk suggest that the officer searches minority drivers more than equally risky white drivers when risk is below

Figure 7: Example where bias is not detected

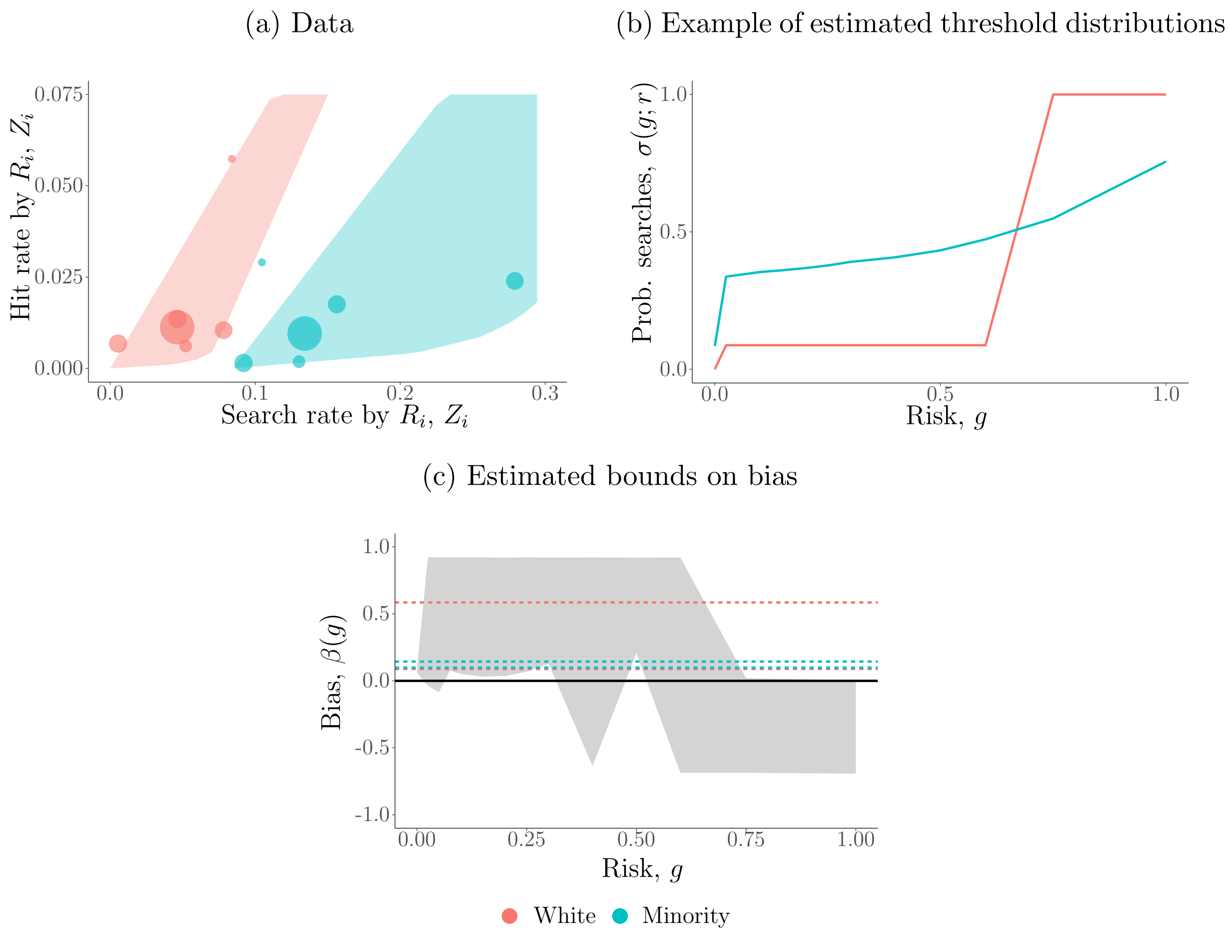


Note: Each dot in the left panel corresponds to the search and hit rates for a particular race and setting. The size of the dots represents the number of stops the data are associated with. Search and hit rates are averaged over the distribution of $X_i \mid R_i = w$. The purple polygon in the left panel represents the data that can be generated by the threshold distribution shown in the right panel.

0.3. However, as risk increases, the bounds on bias decrease and eventually the direction of bias switches. The top right panel shows two distributions of thresholds consistent with the data, and the change in direction of bias occurs where the two lines intersect. Holding their distribution of risk constant, minority drivers are estimated to be searched at least 8.8 percentage points less on average if they were treated as white drivers. In contrast, white drivers are estimated to be searched at least 9.9 percentage points more on average if they were treated as minority drivers. These estimates are large in magnitude considering this officer searches 4.5% of white drivers and 14.6% of minority drivers.

Figure 9 presents the data for another officer who fails the test. This officer searches 4.0% of white drivers and 12.8% of minority drivers. However, this officer has never found contraband on either group of drivers despite having stopped thousands of them and performed hundreds of searches. This result may be interpreted in two ways. The first is that

Figure 8: Example where bias is detected

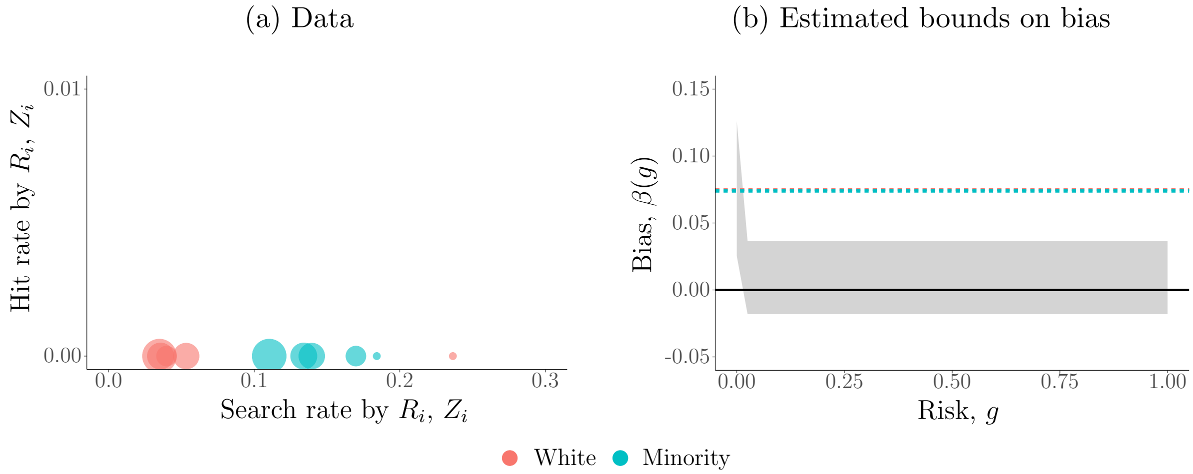


Note: Search and hit rates are averaged over the distribution of $X_i \mid R_i = w$. The red (blue) dashed lines in the bottom panel indicate the bounds on the average bias when the distribution of risk is consistent with that of white (minority) drivers.

the officer has only stopped zero-risk drivers of either race. The racial disparity in search rates is therefore unjustified and bias is detected. Furthermore, if the officer is searching only a fraction of zero-risk drivers from each race, then there is randomness to the search decision, as suggested by Feigenberg and Miller (2022). Such behavior is consistent with the random threshold proposed in (1), but not the fixed thresholds from the earlier literature.

A second and more pessimistic interpretation of this result is that the model is violated. For example, the officer may have been unable to find contraband in all the cases where it

Figure 9: Example of random thresholds



Note: Search and hit rates are averaged over the distribution of $X_i \mid R_i = w$. If white drivers were treated as minority drivers, holding their risk constant, they would be searched approximately 7.5 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 7.4 and 7.5 percentage points less on average.

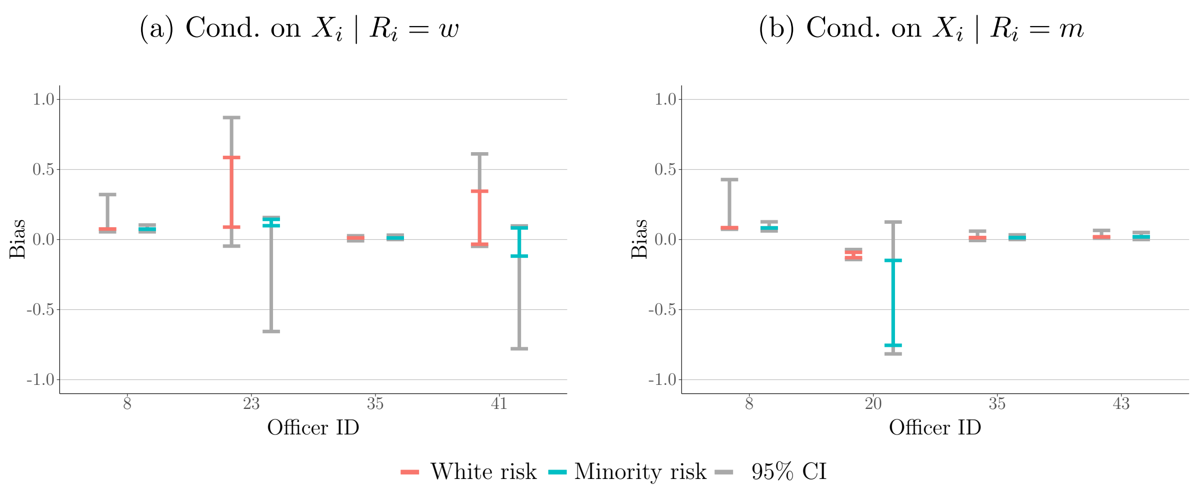
was present, which can be viewed as a violation of Assumption 1(ii).

Figure 10 presents the estimated bounds on the average bias for the officers who fail the test.[42] The left panel corresponds to the estimates where the search and hit rates are averaged over $X_i \mid R_i = w$, and the right panel corresponds to the estimates where the search and hit rates are averaged over $X \mid R_i = m$. The red (blue) bounds correspond to the bias being averaged over the distribution of risk of white (minority) drivers and indicate how much more (less) white (minority) drivers would be searched if they were treated as minorities (whites). The gray bounds correspond to the 95% confidence interval.

When averaging search and hit rates over $X_i \mid R_i = w$, white drivers are estimated to be searched at least 2.4 percentage points more on average (83.1% more relative to the observed search rate of 2.9%) by the biased officers if the drivers were treated as minorities, holding their risk constant. In contrast, minority drivers are estimated to be searched at least 1.2

[42]See Online Appendix F for the estimated bias for all the officers who fail the test.

Figure 10: Bounds on average bias $\mathbb{E}[\beta(G_i)]$ for biased officers



Note: The left (right) panel shows the estimated bounds when search and hit rates are averaged over the distribution of $X_i \mid R_i = w$ ($X_i \mid R_i = m$). Positive average bias indicates minority drivers are searched more often than equally risky white drivers on average. Red (blue) bounds indicate the bias averaged over the distribution of risk for white (minority) drivers in the data. Gray bounds indicate the 95% confidence interval.

percentage points less on average (17.3% less relative to the observed search rate of 6.8%) if they were treated as whites.[43]

When averaging search and hit rates over $X_i \mid R_i = m$, white drivers are estimated to be searched at least 1.2 percentage points more on average (22.3% more relative to the observed search rate of 5.2%) by the biased officers if the drivers were treated as minorities, holding their risk constant. Minority drivers would be searched at most 8.7 percentage points *more* on average (99% more relative to the observed search rate of 8.8%) if they were treated as whites. The potential increase in search rates is because the bounds on the average bias for one officer are $[-0.755, -0.149]$, suggesting he is biased against white drivers on average.[44]

[43] These estimates are obtained by taking a weighted average of the red or blue lower bounds, with the weights being equal to the proportion of stops (conditional on race) made by each officer.

[44] This officer accounts for 15.7% of searches among the biased officers.

For comparison, I apply the test of Knowles et al. (2001) (henceforth, KPT) to these 50 officers in Online Appendix E. In most cases where bias is detected, officers are biased against white drivers. This difference stems from the KPT test compares police department-wide hit rate across race, whereas my method compares both search and hit rates across race for individual officers. Since the KPT test is derived from equilibrium model, it is not intended to evaluate each officer separately and cannot identify which officers are biased.

6 Conclusion

This paper provides a flexible approach to detect and measure racial bias in police traffic searches. The proposed methods are valid amid sample selection on unobservables and statistical discrimination, and circumvents the inframarginality problem. By using an IV to vary the risk among drivers stopped, the methods may be applied to individual officers, allowing for unrestricted heterogeneity in the distributions of thresholds and risk across officers.

This paper also contributes to the literature from a modeling standpoint as earlier papers assumed deterministic thresholds once conditioning on race, whereas I allow the threshold to be random. This relaxation permits a richer notion of bias, where the direction and intensity of bias may depend on the unobserved (to the researcher) risk of the driver. Understanding the relationship between bias and risk can help tailor anti-bias training to mitigate bias against drivers most affected by it. Sharp bounds on the measures of bias are obtained. Implementing these methods involves solving bilinear programs, which are novel in economics.

I apply the methods on police traffic data from the Metropolitan Nashville Police Department and find evidence to suggest 6 of the 50 officers evaluated are biased. For each officer, I estimate bounds on the share of searches stemming from bias. The estimates suggest that the existence and intensity of bias for officers vary with the observable characteristics and unobserved risk of the driver.

References

Aaron, D., K. Mumford, and B. Reese (2019). New Initiative to Further Enhance Public Safety in Nashveill's Entertainment District. *Metropolitan Nashville Police Department Media Release*. https://www.nashville.gov/departments/police/news/new-initiative-further-enhance-public-safety-nashvilles-entertainment (accessed 6/19/2023).

Agan, A. and S. Starr (2018). Ban the Box, Criminal Records, and Racial Discrimination: A Field Experiment. *The Quarterly Journal of Economics 133*(1), 191–235.

Aigner, D. J. and G. G. Cain (1977). Statistical Theories of Discrimination in Labor Markets. *Indutrial and Labor Relations Review 30*(2), 175–187.

Andrews, D. W. and P. Guggenberger (2009). Validity of Subsampling and "Plug-in Asymptotic" Inference for Parameters Defined by Moment Inequalities. *Econometric Theory 25*(3), 669–709.

Anwar, S. and H. Fang (2006). An Alternative Test of Racial Prejudice in Motor Vehicle Searches: Theory and Evidence. *American Economic Review 96*(1), 127–151.

Arnold, D., W. Dobbie, and P. Hull (2022). Measuring Racial Discrimination in Bail Decisions. *American Economic Review 112*(9), 2992–3038.

Arnold, D., W. Dobbie, and C. S. Yang (2018). Racial Bias in Bail Decisions. *The Quarterly Journal of Economics 133*(4), 1885–1932.

Bartlett, R., A. Morse, R. Stanton, and N. Wallace (2022). Consumer-lending Discrimination in the FinTech Era. *Journal of Financial Economics 143*(1), 30–56.

Becker, G. S. (1957). *The Economics of Discrimination*. University of Chicago Press.

Becker, G. S. (1993). Nobel Lecture: The Economic Way of Looking at Behavior. *Journal of Political Economy 101*(3), 385–409.

Bhutta, N. and A. Hizmo (2021). Do Minorities Pay More for Mortgages? *The Review of Financial Studies 34*(2), 763–789.

Bohren, J. A., K. Haggag, A. Imas, and D. G. Pope (2023). Inaccurate Statistical Discrimination: An Identification Problem. *Review of Economics and Statistics*, 1–45.

Bohren, J. A., A. Imas, and M. Rosenberg (2019). The Dynamics of Discrimination: Theory and Evidence. *American Economic Review 109*(10), 3395–3436.

Bugni, F. A., I. A. Canay, and X. Shi (2015). Specification Tests for Partially Identified Models Defined by Moment Inequalities. *Journal of Econometrics 185*(1), 259–282.

Bugni, F. A., I. A. Canay, and X. Shi (2017). Inference for Subvectors and Other Functions of Partially Identified Parameters in Moment Inequality Models. *Quantitative Economics 8*(1), 1–38.

Canay, I. A., M. Mogstad, and J. Mountjoy (2023). On the Use of Outcome Tests for Detecting Bias in Decision Making. *Review of Economic Studies*.

Card, D., A. R. Cardoso, and P. Kline (2016). Bargaining, Sorting, and the Gender Wage Gap: Quantifying the Impact of Firms on the Relative Pay of Women. *The Quarterly journal of economics 131*(2), 633–686.

Chan, D. C., M. Gentzkow, and C. Yu (2022). Selection with Variation in Diagnostic Skill: Evidence from Radiologists. *The Quarterly Journal of Economics 137*(2), 729–783.

Chyn, E., B. Frandsen, and E. Leslie (2025). Examiner and Judge Designs in Economics: A Practitioner's Guide. *Journal of Economic Literature 63*(2), 401–439.

Dobbie, W., A. Liberman, D. Paravisini, and V. Pathania (2021). Measuring Bias in Consumer Lending. *The Review of Economic Studies 88*(6), 2799–2832.

Durlauf, S. N. and J. J. Heckman (2020). An Empirical Analysis of Racial Differences in Police Use of Force: A Comment. *Journal of Political Economy 128*(10), 3998–4002.

Feigenberg, B. and C. Miller (2022). Would Eliminating Racial Disparities in Motor Vehicle Searches Have Efficiency Costs? *The Quarterly Journal of Economics 137*(1), 49–113.

Ferrier, D. (2019). Ferrier Files: Ridgetop Disbands Police Department After Illegal Ticket Quotas Exposed. *Fox 17 WZTV*. https://fox17.com/news/local/ferrier-files-ridgetop-disbands-police-department-after-illegal-ticket-quotas-exposed (accessed 6/19/2023).

Frandsen, B., L. Lefgren, and E. Leslie (2023). Judging Judge Fixed Effects. *American Economic Review 113*(1), 253–77.

Fryer Jr, R. G. (2019). An Empirical Analysis of Racial Differences in Police Use of Force. *Journal of Political Economy 127*(3), 1210–1261.

Gaebler, J., W. Cai, G. Basse, R. Shroff, S. Goel, and J. Hill (2022). A Causal Framework for Observational Studies of Discrimination. *Statistics and public policy 9*(1), 26–48.

Gelbach, J. B. (2021). Testing Economic Models of Discrimination in Criminal Justice. *Social Science Research Network No. 3784953*.

Gelman, A., J. Fagan, and A. Kiss (2007). An Analysis of the New York City Police Department's "Stop-and-frisk" Policy in the Context of Claims of Racial Bias. *Journal of the American Statistical Association 102*(479), 813–823.

Goel, S., J. M. Rao, and R. Shroff (2016a). Personalized Risk Assessments in the Criminal Justice System. *American Economic Review 106*(5), 119–23.

Goel, S., J. M. Rao, and R. Shroff (2016b). Precinct or Prejudice? Understanding Racial Disparities in New York City's Stop-and-Frisk Policy. *The Annals of Applied Statistics 10*(1), 365–394.

Goncalves, F. and S. Mello (2021). A Few Bad Apples? Racial Bias in Policing. *American Economic Review 111*(5), 1406–1441.

Grogger, J. and G. Ridgeway (2006). Testing for Racial Profiling in Traffic Stops from Behind a Veil of Darkness. *Journal of the American Statistical Association 101*(475), 878–887.

Gurobi Optimization, Inc. (2021). Gurobi Optimizer Reference Manual.

Heckman, J. J. and E. Vytlacil (2005). Structural Equations, Treatment Effects, and Econometric Policy Evaluation. *Econometrica 73*(3), 669–738.

Imbens, G. W. and J. D. Angrist (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica 62*(2), 467–475.

Kalinowski, J. J., M. B. Ross, and S. L. Ross (2023). Endogenous Driving Behavior in Tests of Racial Profiling. *Journal of Human Resources*.

King, G. and L. Zeng (2001). Logistic Regression in Rare Events Data. *Political analysis 9*(2), 137–163.

Kleinberg, J., H. Lakkaraju, J. Leskovec, J. Ludwig, and S. Mullainathan (2018). Human Decisions and Machine Predictions. *The Quarterly Journal of Economics 133*(1), 237–293.

Kline, P., E. K. Rose, and C. R. Walters (2022). Systemic Discrimination Among Large US Employers. *The Quarterly Journal of Economics 137*(4), 1963–2036.

Knowles, J., N. Persico, and P. Todd (2001). Racial Bias in Motor Vehicle Searches: Theory and Evidence. *Journal of Political Economy 109*(1), 203–229.

Knox, D., W. Lowe, and J. Mummolo (2020). Administrative Records mask Racially Biased Policing. *American Political Science Review 114*(3), 619–637.

MacDonald, J. M. and J. Fagan (2019). Using Shifts in Deployment and Operations to Test for Racial Bias in Police Stops. In *AEA Papers and Proceedings*, Volume 109, pp. 148–51.

Makofske, M. (2020). Pretextual Traffic Stops and Racial Disparities in their Use. *Working Paper*.

Marx, P. (2022). An Absolute Test of Racial Prejudice. *The Journal of Law, Economics, and Organization 38*(1), 42–91.

McCormick, G. P. (1976). Computability of Global Solutions to Factorable Nonconvex Programs: Part I—Convex Underestimating Problems. *Mathematical programming 10*(1), 147–175.

McDonald, H. (2021). Special Report: MNPD Puts 60 Extra Officers in Nashville's Entertainment District on Weekends. Why? *News Channel 5 Nashville*. https://www.newschannel5.com/news/special-report-mnpd-puts-60-extra-officers-in-nashvilles-entertainment-district-on-weekends-why (accessed 6/19/2023).

Mechoulan, S. and N. Sahuguet (2015). Assessing Racial Disparities in Parole Release. *The Journal of Legal Studies 44*(1), 39–74.

Mehlhorn, K., P. Sanders, and P. Sanders (2008). *Algorithms and Data Structures: The Basic Toolbox*, Volume 55. Springer.

Novak, K. J. and M. B. Chamlin (2012). Racial Threat, Suspicion, and Police Behavior: The Impact of Race and Place in Traffic Enforcement. *Crime & Delinquency 58*(2), 275–300.

Obermeyer, Z., B. Powers, C. Vogeli, and S. Mullainathan (2019). Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science 366*(6464), 447–453.

Persico, N. (2002). Racial Profiling, Fairness, and Effectiveness of Policing. *American Economic Review 92*(5), 1472–1497.

Persico, N. (2009). Racial Profiling? Detecting Bias Using Statistical Evidence. *Annu. Rev. Econ. 1*(1), 229–254.

Persico, N. and P. Todd (2006). Generalising the Hit Rates Test for Racial Bias in Law Enforcement, with an Application to Vehicle Searches in Wichita. *The Economic Journal 116*(515), F351–F367.

Pierson, E., C. Simoiu, J. Overgoor, S. Corbett-Davies, D. Jenson, A. Shoemaker, V. Ramachandran, P. Barghouty, C. Phillips, R. Shroff, et al. (2020). A Large-scale Analysis of Racial Disparities in Police Stops Across the United States. *Nature Human Behaviour 4*(7), 736–745.

Puhr, R., G. Heinze, M. Nold, L. Lusa, and A. Geroldinger (2017). Firth's Logistic Regression with Rare Events: Accurate Effect Estimates and Predictions? *Statistics in medicine 36*(14), 2302–2317.

Rau, N. (2021). In Nashville, Mayor Cooper, Chief Drake Announce Policing Reforms to Address Murders, Gun Crimes. *Tennessee Lookout*. https://tennesseelookout.com/2021/02/01/in-nashville-mayor-cooper-chief-drake-announce-policing-reforms-to-address-murders-gun-crimes/ (accessed 6/19/2023).

Ridgeway, G. (2006). Assessing the Effect of Race Bias in Post-Traffic Stop Outcomes Using Propensity Scores. *Journal of Quantitative Criminology 22*(1), 1–29.

Ridgeway, G. and J. M. MacDonald (2009). Doubly Robust Internal Benchmarking and False Discovery Rates for Detecting Racial Bias in Police Stops. *Journal of the American Statistical Association 104*(486), 661–668.

Roh, S. and M. Robinson (2009). A Geographic Approach to Racial Profiling: The Microanalysis and Macroanalysis of Racial Disparity in Traffic Stops. *Police Quarterly 12*(2), 137–169.

Rose, E. K. (2023). A Constructivist Perspective on Empirical Discrimination Research. *Journal of Economic Literature 61*(3), 906–923.

Sigstad, H. (2026). Monotonicity Among Judges: Evidence from Judicial Panels and Consequences for Judge IV Designs. *American Economic Review 116*(1), 189–208.

Simoiu, C., S. Corbett-Davies, and S. Goel (2017). The Problem of Infra-Marginality in Outcome Tests for Discrimination. *The Annals of Applied Statistics 11*(3), 1193–1216.

Wasserman, M. (2023). Hours Constraints, Occupational Choice, and Gender: Evidence from Medical Residents. *The Review of Economic Studies 90*(3), 1535–1568.